

CHAPTER #4

Topic #1

REDUCTION OF ORDER $\Rightarrow$ Exercise #4.2

$\Rightarrow$ This method is for Homogeneous Differential Equations of 2nd order

$\Rightarrow$ In which one solution is given and other has to be found.

For Example:

$$1. \underbrace{y'' - 4y' + 4y = 0}_{\text{Differential Equation}} ; \boxed{y_1 = e^{2x}} \downarrow \text{Given Solution}$$

Formula for y_2 is

$$\boxed{y_2 = u y_1}$$

where y_1 is a given solution

And u is a function of x formula for u is

$$u = \int \frac{e^{-\int P(x) dx}}{y_1^2} dx \rightarrow \text{It can be taken from given Differential Equation just like in present case}$$

$$y'' - \boxed{4}y' + \cancel{4y} = 0$$

$\downarrow$
 $P(x)$

$\Rightarrow$ And In last you also has to convert equation in standard form

⇒ Here in present case it is already in standard form i.e. $y'' + p(x)y' + q(x)y = 0$

For Example

$$x^2 y'' + 2xy' - 6y = 0 \quad ; \quad y_1 = x^2$$

⇒ This equation is not standard to convert it in standard we have to divide it by co-efficient of y'' here it is x^2

Solved Example:

$$x^2 y'' + 2xy' - 6y = 0$$

÷ by x^2 on b/s

$$y'' + \frac{2}{x}y' - \frac{6}{x^2}y = 0$$

$$p(x) = \frac{2}{x} \quad y_1 = x^2 \quad y_2 = ?$$

$$y_2 = u y_1$$

$$u = \int \frac{-\int p(x) dx}{y_1^2} dx$$

$$u = \int \frac{e^{-\int \frac{2}{x} dx}}{(x^2)^2} dx$$

$$u = \int \frac{e^{-2 \ln x}}{x^4} dx$$

$$u = \int \frac{e^{\ln x^2}}{x^4} dx$$

$$U = \int \frac{x^{-2}}{x^4} dx$$

$$U = \int \frac{1}{x^6} dx$$

$$U = \frac{x^{-6+1}}{-6+1} + C$$

$$U = \frac{x^{-5}}{-5} + C$$

$$y_2 = \left(-\frac{x^{-5}}{5} + C\right)(x^2)$$

$$y = y_1 + y_2$$

$$y = x^2 + \left(-\frac{x^{-5}}{5} + C\right)(x^2)$$

Topic #2

Solution of Homogenous Differential Equations with constant co-efficients (Ex 4.3)

Case I:

⇒ If roots are real and unequal or distinct

$$y_1 = e^{m_1 x} \quad y_2 = e^{m_2 x} \quad \dots \quad y_n = e^{m_n x}$$

$$y_c = y_1 + y_2 + \dots + y_n$$

Case II:

⇒ If roots are real and equal or non-distinct

$$y_1 = e^{mx} \quad \dots \quad y_2 = x e^{mx} \quad \dots \quad y_n = x^{n-1} e^{mx}$$

$$y_c = y_1 + y_2 + \dots + y_n$$

Case III

If roots are Imaginary and unequal or distinct

$$y_c = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

where α = Real Part β = Imaginary Part

STEPS To Solve:

STEP I:

- ⇒ First make Auxiliary Equation
⇒ Method to make Auxiliary Equation is

$$y'' \rightarrow m^2$$

$$y' \rightarrow m$$

$$y \rightarrow 1$$

⋮

$$y^n \rightarrow m^n$$

Example:

$$y'' + 2y' - 6y = 0$$

Auxiliary Eq. is

$$m^2 + 2m - 6 = 0$$

STEP II:

- ⇒ Solve the Auxiliary Equation by quadratic formula
or any method you can also find it directly
by calculator

STEP III:

- ⇒ After Finding Values of m solve it according
to cases above mentioned

example:

$$\rightarrow y''' + 3y'' - 4y' - 12 = 0$$

Auxiliary Equation is

$$m^3 + 3m^2 - 4m - 12 = 0$$

$$m_1 = -3$$

$$m_2 = 2$$

$$m_3 = -2$$

$\Rightarrow$ Here these values are calculated by Calculator

$$y_1 = e^{-3x}$$

$$y_2 = e^{2x}$$

$$y_3 = e^{-2x}$$

$\rightarrow$ By case I

$$y_c = c_1 e^{-3x} + c_2 e^{2x} + c_3 e^{-2x}$$

$$\rightarrow 3y'' + y = 0$$

Auxiliary Equation is

$$3m^2 + 1 = 0$$

$$m^2 = -\frac{1}{3}$$

$$m = \pm \frac{1}{\sqrt{3}} i$$

Here $\alpha = 0$ $\beta = \frac{1}{\sqrt{3}}$ roots are imaginary so By case III

$$y_c = e^{0x} (c_1 \cos \frac{1}{\sqrt{3}} x + c_2 \sin \frac{1}{\sqrt{3}} x)$$

$$\rightarrow y'' - 10y' + 25y = 0$$

Auxiliary Eq. is

$$m^2 - 10m + 25 = 0$$

$$m^2 - 5m - 5m + 25 = 0$$

$$m(m-5) - 5(m-5) = 0$$

$$(m-5)(m-5) = 0$$

$$m_1 = 5$$

$$m_2 = 5$$

As roots are equal so, by case II

$$y_1 = e^{5x}$$

$$y_2 = xe^{5x}$$

$$y_c = c_1 e^{5x} + c_2 x e^{5x}$$

Topic #3

Solution of Non-Homogeneous Equations By

UC-METHOD (Ex 4.4)

$\Rightarrow$ The main difference between Homogeneous and non-Homogeneous equation is on the other side in Homogeneous there is always 0 but in non there can be any function.

$\Rightarrow$ STEPS To Solve

$\rightarrow$ First find y_c (Characteristic solution) by let Non-Homogeneous Equation equal to 0

$\rightarrow$ When it become equal to 0 then only apply

three cases as earlier discussed.

→ Then And y_p picture is Given

→ Then take derivatives only that are required to given eq.

Example:

$$y'' - 8y' + 20y = 100x^2 - 26xe^x$$

Let

$$y'' - 8y' + 20y = 0$$

Auxiliary Eq. is

$$m^2 - 8m + 20 = 0$$

$$m = 4 \pm 2i$$

$$y_c = e^{4x}(C_1 \sin 2x + C_2 \cos 2x)$$

Now $y_p = (Ax^2 + Bx + C) + (Dx + E)e^x$

$$y_p' = (2Ax + B) + (Dx + E)e^x + (D)e^x$$

$$y_p'' = 2A + (Dx + E)e^x + (D)e^x + De^x$$

$$y_p'' = 2A + (Dx + E)e^x + 2De^x$$

So, only two derivatives are taken as there are two in question put this in Original eq.

TABLE 4.4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. $x^2 e^{5x}$	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

$$2A + (Dx + E)e^x + 2De^x - 8((2Ax + B) + (Dx + E)e^x + (D)e^x) + 20(($$

$$(Ax^2 + Bx + C) + (Dx + E)e^x = 100x^2 - 26xe^x$$

→ Here After comparing Find A, B, C, D, E find y_p
 putting values then find y by $y = y_c + y_p$

→ In some cases you has to be multiply x by y_p if y_c and y_p has same values Just like

$$y'' + y = 2x \sin x$$

Auxiliary Eq. is

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m = \pm i$$

$$y_c = \boxed{C_1 \cos x} + \boxed{C_2 \sin x}$$

$$y_p = (Ax + B) \sin x + (Cx + D) \cos x$$

$$y_p = Ax \sin x + \boxed{B \sin x} + Cx \cos x + \boxed{D \cos x}$$

As you see enboxed are same so multiply by x in y_p

$$y_p = Ax^2 \sin x + Bx \sin x + Cx^2 \sin x + Dx \cos x$$

⇒ Here atleast one term has to be same to multiply by x

Topic # 4

VARIATION OF PARAMETERS (Ex 1.6)

- All question ~~can't~~ of UC-Method can be solve by variation of parameters but variation of parameters question can't be by UC-Method.
- Why we get use of variation of Parameters ?
Because when functions like $\sec x$, $\ln x$ rather than $\sin$, $\cos$, linear, quadratic, cubic or exponential comes then we can't apply UC. Because the derivative of function like $\sec x$ is $\sec x \tan x$ it increases not reduces.

STEPS:

STEP I:

→ Find y_c

STEP II:

→ Let y_p

For Example:

$$x^2 y'' - 2y = 0$$

$$y'' + y = \sec \theta \tan \theta$$

Let

$$y'' + y = 0$$

Auxiliary Equation

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m = \pm i$$

$$y_c = C_1 \cos x + C_2 \sin x$$

→ Here after finding y_c let y_p by changing C_1 to U_1 and C_2 to U_2 and so on.

let

$$y_p = U_1 \cos x + U_2 \sin x$$

→ Here U_1 and U_2 are functions of x where $\cos x = y_1$ & $\sin x = y_2$

To FIND U_1 and U_2 .

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Here y_1 and y_2 for example if y_3 is also present then

$$W \text{ will be } \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

→ After finding W if ~~you want~~ you have to check whether your equation is in standard form or not i.e. $y'' + P(x)y' + Q(x)y = 0$

In present case

$y'' + y = \sec \theta \tan \theta$ is already in standard form

Other:

$$3y'' + y' + 6y = e^x \sec x$$

Here you have to convert so $\div$ by 3

$$y'' + \frac{y'}{3} + 2y = \frac{e^x \sec x}{3}$$

After This write W_1, W_2

$$W_1 = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

what is $f(x)$?

$f(x)$ will always be the function that is on the other side of equal in eq.

$$\rightarrow y'' + y = \boxed{\sec \theta \tan \theta} \rightarrow f(x)$$

$$y'' + \frac{y'}{3} + 2y = \boxed{\frac{e^x \sec x}{3}} \rightarrow f(x)$$

After that

$$U_1' = \frac{W_1}{W}$$

$$U_2' = \frac{W_2}{W}$$

→ Integrate U_1' and U_2' you will get U_1 and U_2 respectively.

→ put U_1 and U_2 in $y_p = U_1 y_1 + U_2 y_2$ you will get y_p and then put y_c and y_p in $y = y_c + y_p$ you'll get the solution.

Solved Example:

→ $y''' + y' = \tan x$

Let

$$y''' + y' = 0$$

Auxiliary Eq. is

$$m^3 + m = 0$$

$$m(m^2 + 1) = 0$$

$$m = 0$$

$$m^2 = -1$$

$$\boxed{m = 0}$$

$$\boxed{m = \pm i}$$

~~$y_1 = e^{0x}$ $y_2 = C_1 \cos x + C_2 \sin x$~~

$$y_c = C_1 \cos x + C_2 \sin x + C_3 e^{0x}$$

$$y_c = C_1 \cos x + C_2 \sin x + C_3$$

where

$$y_1 = \cos x \quad y_2 = \sin x \quad y_3 = 1$$

Let $y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

$$W = \begin{vmatrix} \cos x & \sin x & 1 \\ -\sin x & \cos x & 0 \\ -\cos x & -\sin x & 0 \end{vmatrix}$$

$$W = \cos x (0 - 0) - \sin x (0 - 0) + 1 (\sin^2 x + \cos^2 x)$$

$$\boxed{W = 1}$$

$$y''' + y' = \boxed{\tan x} \rightarrow f(x)$$

Here the equation is already in standard form

So,

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ f(x) & y_2'' & y_3'' \end{vmatrix}$$

$$W_1 = \begin{vmatrix} 0 & \sin x & 1 \\ 0 & \cos x & 0 \\ \tan x & -\sin x & 0 \end{vmatrix}$$

$$W_1 = 0 - \sin x (0) + 1 (\cos x \tan x)$$

$$W_1 = \cos x \tan x$$

$$W_2 = \begin{vmatrix} \cos x & 0 & 1 \\ -\sin x & 0 & 0 \\ -\cos x & \tan x & 0 \end{vmatrix}$$

$$W_2 = \cos x(0) - 0 + 1(-\sin x \tan x)$$

$$W_2 = -\sin x \tan x$$

$$W_3 = \begin{vmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ -\cos x & -\sin x & \tan x \end{vmatrix}$$

$$W_3 = \cos x(\cos x \tan x) - \sin x(-\sin x \tan x) + 0$$

$$W_3 = \cos^2 x \tan x + \sin^2 x \tan x$$

$$W_3 = \tan x$$

$$\left. \begin{aligned} u_1' &= \frac{W_1}{W} \\ u_2' &= \frac{W_2}{W} \\ u_3' &= \frac{W_3}{W} \end{aligned} \right\} \text{Integrate all then put in yp}$$